

Gen AI Platform for Automated Content Transformation

Architecture — one source, many deliverables, fully traceable

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NTRO

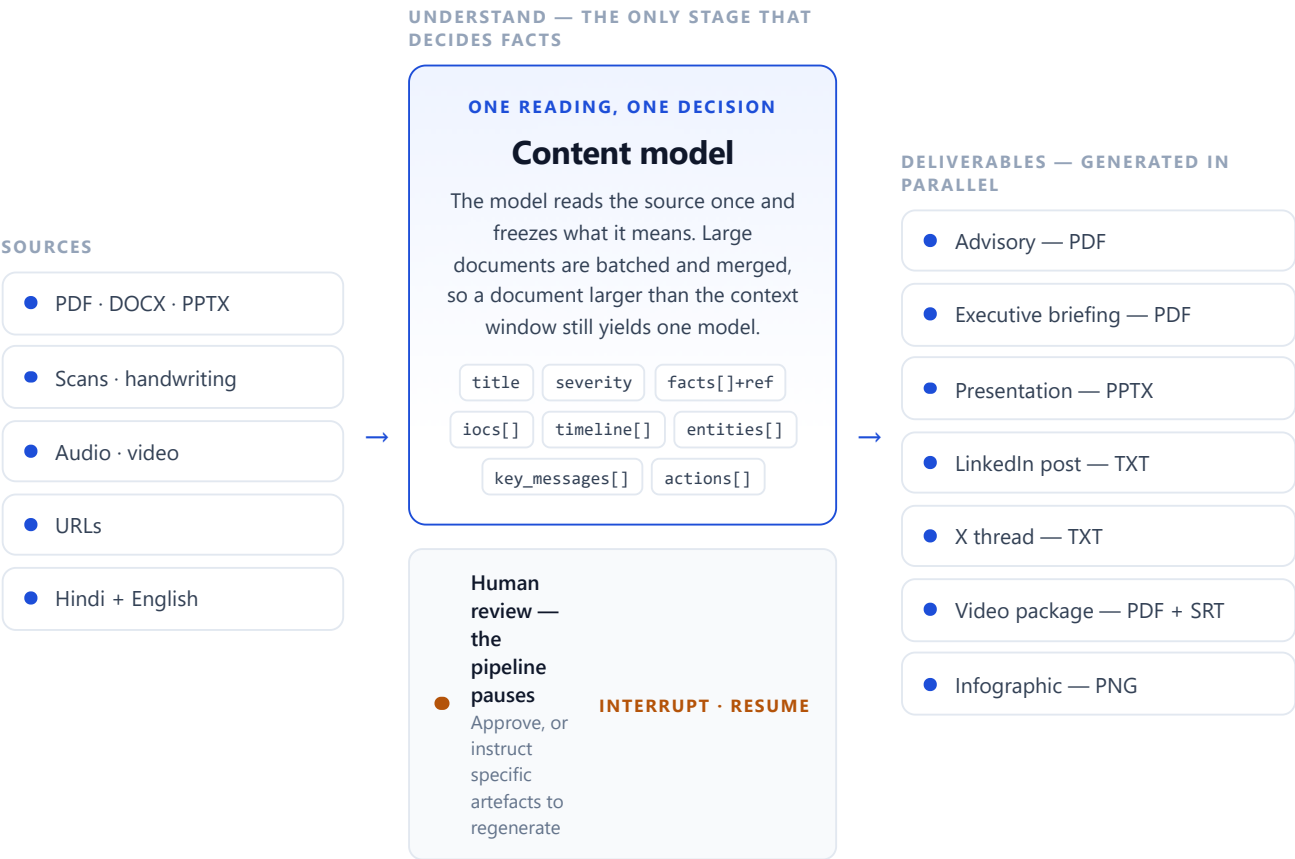
BLOCKCHAIN & CYBERSECURITY

1 · THE ONE DECISION THAT SHAPES EVERYTHING

The obvious implementation prompts the model once per desired output. It fails in four ways: **fact drift** (each call re-reads the source and resamples its facts), **inconsistency** (seven artefacts become seven interpretations), **no traceability** (prose cannot be pointed back at a source passage), and **cost** (the full source is re-sent for every output).

This platform instead **separates understanding from expression**. The source is read exactly once and frozen into a single structured **content model**; every artefact is a projection of that one object rather than a fresh reading. That single decision is what makes multi-output consistent rather than merely parallel, and it is why no two deliverables can disagree.

2 · ARCHITECTURE



3 · COMPONENTS

MODULE	RESPONSIBILITY
engine/graph.py	LangGraph state machine: ingest → clean → understand → parallel fan-out → human review → render or refine
engine/ingestion/	Ten modules turning every input type into referenced text blocks. Docling for documents, vision OCR for thin text and figures, transcription for media, translation to English
engine/understand.py	Builds the content model; batches and merges oversized sources; grounds every fact against the corpus and drops what it cannot trace
engine/generate.py	Drafts one artefact per branch, then verifies it (see §5) and regenerates once if grounding is poor

4 · FOUR INVARIANTS

Every design decision below follows from these. They are the properties a reviewer should test us against.

- 1

Every block carries a reference from birth. Refs are assigned at parse time and never rebuilt downstream. This is the precondition for citation tracing.
- 2

One call decides facts; the rest only express them. The writer never sees the raw source, so it cannot reinterpret it — only render what understanding already decided.
- 3

The model produces blueprints; libraries produce files. Same draft renders byte-identically, which is what makes the SHA-256 export manifest meaningful and the output auditable.
- 4

Human review is structural. The graph pauses via a checkpointed interrupt; it is not a UI that hides output, and the run survives a restart of the process.

5 · VERIFICATION AND HUMAN REVIEW

Verification is an input to generation, not a report: a poor score triggers a rewrite before a human ever sees the draft. Three independent checks catch different error classes.

CHECK	CATCHES
Lexical grounding	Sentences with no support in the corpus. Stopwords are excluded, so the threshold cannot be met by grammar alone.
Number verification	Changed quantities, dates and versions — the failure a wording comparison can never see. A source stating 72 hours no longer approves a draft saying 9 hours .
Indicator integrity	Fabricated IOCs. Any IP, hash, URL or CVE in the output that is absent from the verified set is flagged.

Each claim is returned with a verdict and its `source_ref`. Review is a genuine gate: the graph stops, and the operator **accepts** or **refines** — refining regenerates only the affected artefacts, not the whole run.

6 · DEPLOYMENT, AND INDEPENDENCE FROM ANY ONE MODEL

Every model call passes through a provider interface, so the platform is not tied to a vendor. A configured chain is tried in order and rotated **per call**, with cooldowns and capability routing — an image call never goes to a text-only provider.

Because quota is per account rather than per model, rotation is the only mechanism that survives a provider outage. The same interface is what allows the system to run entirely offline.

Cloud	Gemini · Groq · OpenRouter, with automatic failover
On-premises	Local Ollama — no external calls, no data leaves the network
Offline	Deterministic stub, for development and for demonstrating without any key

7 · STACK AND MEASURED BEHAVIOUR

CONCERN	CHOICE	MEASURED	RESULT
Orchestration	LangGraph state machine, checkpointed	Endpoint to review	45–90 s for a 2-page source and 3 artefacts
Documents	Dodling, with layout and table structure	Handwritten scan	20 pages read with correct mathematical notation
OCR	Vision model, batched for rate limits	Image-heavy PDF	17 → 11,406 characters after a page-selection fix
Media	Provider transcription	803-page document	Answered in 13 s via a fast text path
Rendering	python-pptx · reportlab · Pillow	Test suite	82 offline tests, no network required
Service	FastAPI · Next.js		

Why this is more than a wrapper. The engine runs on hostile real-world input — scanned pages, handwritten notes, Hindi audio, sixty-page decks — and produces real files (PDF, PPTX, PNG, SRT) rather than text. Understanding happens once, verification happens automatically, and a human approves before anything is delivered.